

Fixed Partitioning :

Multi-programming with fixed partitioning is a contiguous memory management technique in which the main memory is divided into partitions of equal or different sizes. The operating system always resides in the first partition while the other partitions can be used to store user processes. The memory is assigned to the processes in contiguous way.

Multiprogramming with Fixed partitions

- This method allows multiple processes to execute simultaneously.
- Here memory is divided into fixed sized partitions. Size can be equal or unequal for different partitions.
- Generally unequal partitions are used for better utilizations.
- Each partition can accommodate exactly one process, means only single process can be placed in one partition.
- The partition boundaries are not movable.
- Whenever any program needs to be loaded in memory, a free partition big enough to hold the program is found. This partition will be allocated to that program or process.
- If there is no free partition available of required size, then the process needs to wait. Such process will be put in a queue.
- There are two ways to maintain queue
 - Using multiple Input Queues.
 - Using single Input Queue.

Advantages of Fixed Partitioning –

1. Easy to implement:
Algorithms needed to implement Fixed Partitioning are easy to implement. It simply requires putting a process into a certain partition without focusing on the emergence of Internal and External Fragmentation.
2. Little OS overhead:
Processing of Fixed Partitioning requires lesser excess and indirect computational power.

Disadvantages of Fixed Partitioning –

1. Internal Fragmentation:
Main memory use is inefficient. Any program, no matter how small, occupies an entire partition. This can cause internal fragmentation.
2. External Fragmentation:
The total unused space (as stated above) of various partitions cannot be used to load the
3. processes even though there is space available but not in the contiguous form (as spanning is not allowed).
4. Limit process size:
Process of size greater than the size of the partition in Main Memory cannot be accommodated. The partition size cannot be varied according to the size of the incoming process's size. Hence, the process size of 32MB in above-stated example is invalid.
5. Limitation on Degree of Multi-programming:
Partitions in Main Memory is made before execution or during system configure. Main Memory is divided into a fixed number of partitions.